

Endpoint Forensics and Threat Hunting Lab Using Velociraptor

Project Overview

This project demonstrates the deployment and use of Velociraptor for endpoint visibility, digital forensics, and threat hunting. A Windows 11 endpoint was connected to a Velociraptor server hosted on Ubuntu Linux to collect and analyze endpoint telemetry.

The objective was to establish a baseline of normal system activity and identify potential indicators of compromise (IOCs) through process, network, service, and persistence analysis.

Environment

Server

- Ubuntu Linux
- Velociraptor Server v0.77.1
- IP Address: 192.168.X.XXX

Endpoint

- Microsoft Windows 11 Home
- Hostname: Edward
- Velociraptor Agent v0.77.1
- IP Address: 192.168.X.XXX

Additional Security Tools

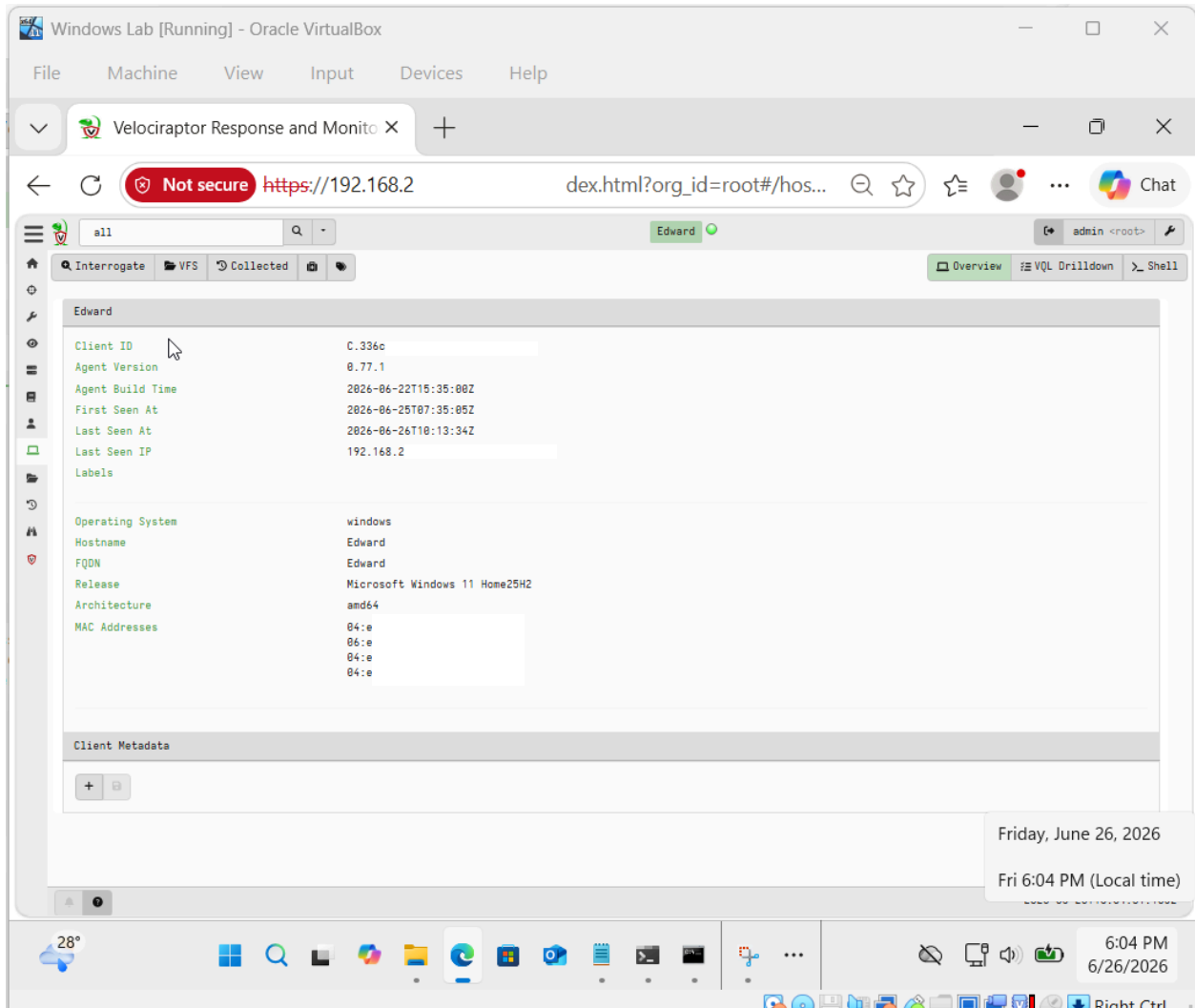
- Wazuh SIEM
- Suricata IDS

Objectives

1. Deploy and configure a Velociraptor server.
2. Connect a Windows endpoint to the server.
3. Collect forensic artifacts from the endpoint.
4. Establish a baseline of normal system activity.
5. Perform basic threat hunting and anomaly detection.

Methodology

1. Client Enrollment



The Windows endpoint was successfully enrolled into Velociraptor and verified through the client information page.

Collected information included:

- Hostname
- Operating System Version
- Agent Version
- Network Information
- Hardware Architecture

2. Process Analysis

State	FlowId	Artifacts	Created	Last Active	Creator	Mb	Rows
✓	F.DBUEACNB2PC8I	Windows.System.Pslist	2026-06-25T08:22:10.117Z	2026-06-25T08:22:20.387Z	admin	0 b	290
✓	F.D8UDKAAPIC1H2	Generic.Client.Info	2026-06-25T07:35:05.630Z	2026-06-25T07:35:05.697Z	InterrogationService	0 b	26

CreateTime	Pid	Ppid	TokenIsElevated	Name	CommandLine	Exe	TokenInfo	Hash	AuthenticCode	Username	WorkingSetSize
2026-06-18T02:48:32.074Z	4	0	true	System						NT AUTHORITY\SYSTEM	147456
2026-06-18T02:48:28.528Z	176	4	false	Registry						NT AUTHORITY\SYSTEM	23572480
2026-06-18T02:48:32.078Z	632	4	true	smss.exe	\SystemRoot\System32\smss.exe	C:\Windows\System32\smss.exe		{ "MD5": "d997898238c9ffde81d6d3aef879e1", "SHA1": "bd934d7c1885852331a"	{ "Filename": "C:\\Windows\\System32\\smss.exe", "ProgramName": "Microsoft Windows", "PublisherLink": null, "MoreInfoLink": null }	NT AUTHORITY\SYSTEM	167936

Artifact Used:

- Windows.System.Pslist

Purpose:

- Enumerate all running processes.
- Identify suspicious executables.
- Detect process masquerading attempts.

Findings:

- 290 active processes were identified.
- Core Windows processes such as explorer.exe, svchost.exe, lsass.exe, winlogon.exe, and services.exe were present.
- No suspicious process names or unusual executable paths were identified.

Assessment:

- No evidence of malicious process execution was observed.

3. Network Analysis

The screenshot shows a web application interface for network analysis. The top part displays a table of artifacts, and the bottom part shows a detailed view of the 'Windows.Network.Netstat' artifact.

State	FlowId	Artifacts	Created	Last Active	Creator	Mb	Rows
✓	F.D8UEDD835JQCE	Windows.Network.Netstat	2026-06-25T08:28:36.020Z	2026-06-25T08:28:36.705Z	admin	0 b	230
✓	F.D8UEACN82PC8I	Windows.System.Pslist	2026-06-25T08:22:19.117Z	2026-06-25T08:22:20.387Z	admin	0 b	290
✓	F.D8UDKAAPIC1H2	Generic.Client.Info	2026-06-25T07:35:05.630Z	2026-06-25T07:35:05.697Z	InterrogationService	0 b	26

Pid	Name	Family	Type	Status	Laddr.IP	Laddr.Port	Raddr.IP	Raddr.Port	Timestamp
1420	svchost.exe	IPv4	TCP	LISTEN	0.0.0.0	135	0.0.0.0	0	2026-06-18T02:48:37Z
4	System	IPv4	TCP	LISTEN	192.1	139	0.0.0.0	0	2026-06-25T06:21:14Z
8784	svchost.exe	IPv4	TCP	LISTEN	0.0.0.0	5040	0.0.0.0	0	2026-06-25T07:01:44Z
10216	Spotify.exe	IPv4	TCP	LISTEN	127.0.0.1	7768	0.0.0.0	0	2026-06-24T07:49:25Z
4048	AsusSoftwareManager.exe	IPv4	TCP	LISTEN	127.0.0.1	24830	0.0.0.0	0	2026-06-18T02:48:43Z
2636	AppleMobileDeviceService.exe	IPv4	TCP	LISTEN	127.0.0.1	27015	0.0.0.0	0	2026-06-18T02:48:40Z
4	System	IPv4	TCP	LISTEN	127.0.0.1	28385	0.0.0.0	0	2026-06-18T02:48:44Z
4	System	IPv4	TCP	LISTEN	127.0.0.1	28390	0.0.0.0	0	2026-06-18T02:48:44Z
24436	chrome.exe	IPv4	TCP	ESTAB	192.1	49200	74.125	443	2026-06-25T08:25:41Z
24436	chrome.exe	IPv4	TCP	ESTAB	192.1	49289	57.144	443	2026-06-25T07:16:17Z

Artifact Used:

- Windows.Network.Netstat

Purpose:

- Identify active network connections.
- Detect unauthorized outbound communication.
- Review listening services.

Findings:

- Network activity was consistent with normal operating system and user activity.
- No suspicious remote connections were identified.

Assessment:

- No indicators of command-and-control (C2) communications were observed.

4. Service Analysis

The screenshot shows a web-based security tool interface. At the top, there's a browser address bar with 'https://192.168.1.100' and a search bar. Below the browser, there's a navigation bar with 'all' and 'Edward Connected'. The main content area displays a table of artifacts. The table has columns: State, FlowId, Artifacts, Created, Last Active, Creator, Mb, and Rows. The artifacts listed are: Windows.System.Services, Windows.Network.Netstat, Windows.System.Pslist, and Generic.Client.Info. Below the artifacts table, there's a tabbed interface with 'Artifact Collection', 'Uploaded Files', 'Requests', 'Results', 'Log', and 'Notebook'. The 'Results' tab is active, showing a detailed view of Windows services. The detailed view has columns: State, Name, DisplayName, Status, Pid, ExitCode, StartMode, PathName, ServiceType, UserAccount, Created, and ServiceDll. The services listed are: ADPSvc (Stopped, Aggregated Data Platform Service, OK, 0, 1077, Manual, C:\WINDOWS\system32\svchost.exe -k LocalServiceAuthentication -p) and ALG (Stopped, Application Layer Gateway Service, OK, 0, 1077, Manual, C:\WINDOWS\system32\alg.exe, Own Process, NT AUTHORITY\LocalService, 2024-04-01T07:26:32.631Z, C:\WINDOWS\system32\alg.exe).

State	FlowId	Artifacts	Created	Last Active	Creator	Mb	Rows
✓	F.DBUEFGN22624	Windows.System.Services	2026-06-25T08:33:06.504Z	2026-06-25T08:33:08.557Z	admin	0 b	298
✓	F.DBUEDD0J5JQCE	Windows.Network.Netstat	2026-06-25T08:28:36.020Z	2026-06-25T08:28:36.705Z	admin	0 b	230
✓	F.DBUEACN82PC8I	Windows.System.Pslist	2026-06-25T08:22:10.117Z	2026-06-25T08:22:20.387Z	admin	0 b	298
✓	F.DBUDKAAPIC1H2	Generic.Client.Info	2026-06-25T07:35:05.630Z	2026-06-25T07:35:05.697Z	InterrogationService	0 b	26

State	Name	DisplayName	Status	Pid	ExitCode	StartMode	PathName	ServiceType	UserAccount	Created	ServiceDll
Stopped	ADPSvc	Aggregated Data Platform Service	OK	0	1077	Manual	C:\WINDOWS\system32\svchost.exe -k LocalServiceAuthentication -p	Share Process	NT AUTHORITY\LocalService	2026-04-11T09:46:12.990Z	C:\WINDOWS\system32\adpsvc.dll
Stopped	ALG	Application Layer Gateway Service	OK	0	1077	Manual	C:\WINDOWS\system32\alg.exe	Own Process	NT AUTHORITY\LocalService	2024-04-01T07:26:32.631Z	

Artifact Used:

- Windows.System.Services

Purpose:

- Enumerate running Windows services.
- Identify unauthorized or suspicious services.

Findings:

- Running services corresponded to standard Windows functionality.
- Service executable paths appeared legitimate.

Assessment:

- No unauthorized services were detected.

5. Persistence Analysis

The screenshot shows a web application interface with a top navigation bar and a main content area. The top bar includes a search bar with the text 'all', a status indicator 'Edward Connected', and a user profile 'admin <root>'. The main content area displays a table of artifacts with columns: State, FlowId, Artifacts, Created, Last Active, Creator, Mb, and Rows. Below this table, there are tabs for 'Artifact Collection', 'Uploaded Files', 'Requests', 'Results', 'Log', and 'Notebook'. The 'Results' tab is selected, showing a detailed view of a task named '\ASUS Update Checker 2.0'.

State	FlowId	Artifacts	Created	Last Active	Creator	Mb	Rows
✓	F.D8UEGD1NV3IHA	Windows.System.TaskScheduler	2026-06-25T08:35:00.035Z	2026-06-25T08:35:06.228Z	admin	0 b	282
✓	F.D8UEFGN22624	Windows.System.Services	2026-06-25T08:33:06.504Z	2026-06-25T08:33:08.557Z	admin	0 b	298
✓	F.D8UEEDBJ5JQCE	Windows.Network.Netstat	2026-06-25T08:28:36.020Z	2026-06-25T08:28:36.705Z	admin	0 b	238
✓	F.D8UEACNB2PC8I	Windows.System.Pslist	2026-06-25T08:22:10.117Z	2026-06-25T08:22:20.387Z	admin	0 b	290

TaskName	Mtime	Command	Arguments	UserId	RunLevel	LogonType	StateChange	ComHandler	RegistrationTime	StartBoundary
\ASUS Update Checker 2.0	2026-04-10T18:00:56.508Z	c:\windows\system32\driverstore\filerep\ository\asussci.inf_and64_4bff547a9eb4deccl\asussoftw		S-1-5-18	HighestAvailab	le				

Artifact Used:

- Windows.System.TaskScheduler

Purpose:

- Identify persistence mechanisms commonly used by attackers.
- Review scheduled tasks for unauthorized activity.

Findings:

- Scheduled tasks were associated with legitimate Windows maintenance and update operations.
- No suspicious persistence mechanisms were observed.

Assessment:

- No malicious persistence techniques were identified.

Threat Hunting Observations

The endpoint was examined for common indicators of compromise including:

- Suspicious process names
- Processes executing from temporary directories
- Unauthorized network connections
- Unknown Windows services
- Malicious scheduled tasks

No indicators of compromise were identified during the baseline assessment.

Skills Demonstrated

- Endpoint Forensics
- Digital Forensics and Incident Response (DFIR)
- Threat Hunting
- Windows Artifact Analysis
- Endpoint Visibility
- Security Monitoring
- Velociraptor Administration
- Security Operations Center (SOC) Operations

Results

The Windows 11 endpoint was successfully investigated using Velociraptor. Multiple forensic artifacts were collected and analyzed to establish a baseline of normal activity.

The investigation found no evidence of malicious processes, unauthorized services, suspicious network connections, or persistence mechanisms. This baseline can be used for future comparisons during incident response and threat hunting exercises.

Conclusion

This project demonstrates the practical use of Velociraptor as an endpoint visibility and forensic investigation platform. Through the collection and analysis of endpoint artifacts, a baseline of

system activity was established, providing a foundation for future threat hunting and incident response activities.